



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE-364: Software Development Project - I

User Manual Document

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AmarVote : Secure Election Monitoring And Management System

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Contents

1	Introduction	2
1.1	Purpose of the Manual	2
1.2	Brief Overview of the Web Application	2
2	System Requirements	3
2.1	Supported Browsers and Devices	3
2.2	Internet Requirements	4
2.3	Required Accounts and Permissions	4
3	Getting Started	4
3.1	Sign-up / Login Process	4
3.2	Logout Process	5
4	User Roles	6
4.1	Role Hierarchy and Permissions Matrix	6
4.1.1	BEC Admin (Administrator)	6
4.1.2	Presiding Officer	6
4.1.3	Law Enforcement (Police)	7
5	Features and Functionality	7
5.1	BEC Admin Features	7
5.1.1	User Management	7
5.1.2	Vote Analytics	8
5.1.3	Incident Oversight	9
5.2	Presiding Officer Features	10
5.2.1	Vote Submission	10
5.2.2	Vote Correction Request	11
5.2.3	Incident Reporting	12
5.3	Law Enforcement Features	12
5.3.1	Incident Monitoring	12
5.3.2	Incident Response	13
5.3.3	Interactive Map	14
6	Navigation Guide	15
6.1	Menu Structure and Dashboards	15
6.2	Page Descriptions	15
6.3	Buttons, Icons, and Special Elements	16
7	Error Handling and Messages	16
7.1	Common Errors and Causes	16
7.2	Error Messages	16
7.3	How to Fix Common Issues	17
7.3.1	Login Issues	17
7.3.2	Performance Issues	17
7.3.3	Data Sync Issues	17
7.3.4	Mobile Device Issues	17

1 Introduction

1.1 Purpose of the Manual

This User Manual Report provides comprehensive documentation for AmarVote, a secure web-based Election Monitoring and Management System designed for the Bangladesh Election Commission (BEC). The manual serves as the definitive guide for all authorized users to effectively operate the system during electoral processes.

Manual Objectives:

- Provide step-by-step operational procedures for all user roles
- Ensure proper understanding of system capabilities and limitations
- Guide users through standard workflows and best practices
- Document troubleshooting procedures and error resolution
- Maintain compliance with electoral security protocols

1.2 Brief Overview of the Web Application

AmarVote is a sophisticated web-based election monitoring system designed to support transparent, secure, and efficient electoral processes in Bangladesh. The platform integrates real-time data collection, incident management, and comprehensive reporting capabilities.

Core Application Features:

- **Real-time Monitoring Dashboard:** Live election data visualization and analytics
- **Secure Vote Management:** Encrypted vote submission with correction workflows
- **Incident Tracking System:** Comprehensive incident reporting with GPS tracking
- **Role-Based Security:** Multi-level access control with JWT authentication
- **Audit Trail System:** Complete activity logging for accountability
- **Mobile-Responsive Interface:** Cross-device compatibility for field operations

Technology Architecture:

- **Frontend:** Next.js 16.1 Progressive Web Application with React 18.3 and TypeScript
- **Backend:** Next.js API Routes (server-side) implementing RESTful services
- **Database:** MongoDB Atlas with Mongoose ODM and encrypted data storage
- **Security:** JWT-based authentication with rate limiting, RBAC enforcement, and audit logging
- **Deployment:** Cloud-hosted PWA accessible via modern browsers with high availability

2 System Requirements

2.1 Supported Browsers and Devices

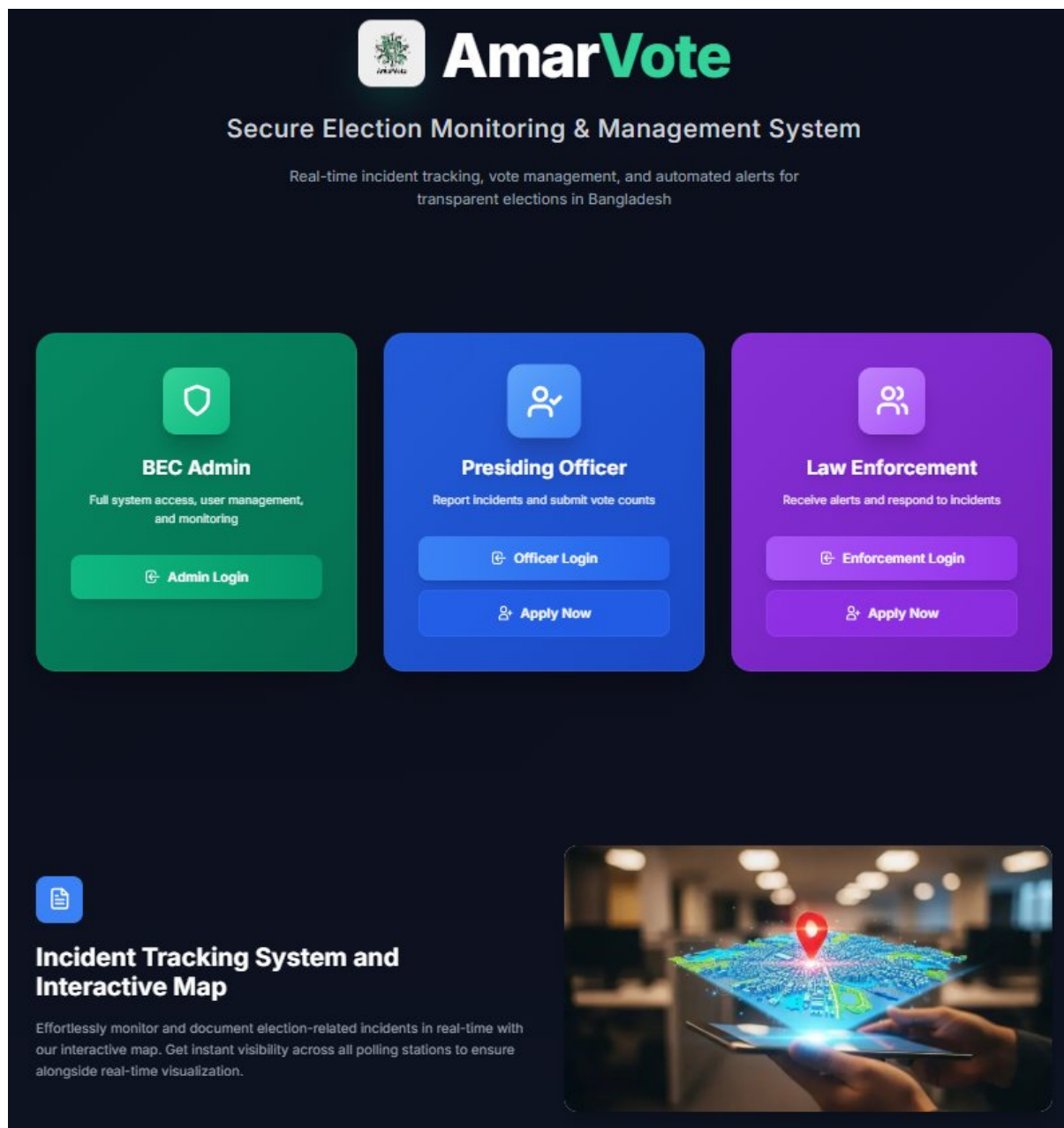


Figure 1: AmarVote – System Interface Preview

Recommended Browsers (Fully Supported):

Browser	Version	Notes
Google Chrome	90+	Recommended for optimal performance
Mozilla Firefox	88+	Full feature support
Microsoft Edge	90+	Compatible with all functions
Safari (macOS)	14+	iOS Safari 14+ also supported

Device Compatibility:

- Desktop/Laptop: Windows 10+, macOS 10.15+, Linux Ubuntu 18.04+
- Tablet: iPad (iOS 14+), Android tablets (Android 10+)
- Mobile: iPhone (iOS 14+), Android phones (Android 10+)
- Screen Resolution: Minimum 1280×720, Recommended 1366×768+

Unsupported Browsers:

- Internet Explorer (all versions)
- Chrome versions below 80
- Firefox versions below 75
- Safari versions below 13

2.2 Internet Requirements

Connection Specifications:

- Minimum Speed: 1 Mbps download, 512 Kbps upload
- Recommended Speed: 5 Mbps download, 2 Mbps upload
- Latency: Maximum 200ms for optimal performance
- Stability: Continuous connection required during vote submission

Network Requirements:

- HTTPS protocol support (TLS 1.2 or higher)
- WebSocket support for real-time updates
- JavaScript enabled
- Cookies enabled for session management
- Local storage support for offline capabilities

2.3 Required Accounts and Permissions

Account Prerequisites:

- Valid BEC-issued credentials (username and password)
- Assigned user role (BEC Admin, Presiding Officer, or Law Enforcement)
- Active account status (not suspended or disabled)
- Completed user verification process as defined by BEC

System Permissions:

- Location services access (for GPS-based incident reporting)
- Camera access (for incident photo uploads)
- File upload permissions (for document attachments)

3 Getting Started

3.1 Sign-up / Login Process

Initial Access (First-Time Users):

1. **Receive Credentials:** Obtain username and temporary password from a BEC Administrator
2. **Access System:** Navigate to official AmarVote URL (provided by admin)

3. **Role Selection:** Click appropriate role card on homepage:
 - BEC Admin – Green card for administrators
 - Presiding Officer – Blue card for polling officers
 - Law Enforcement – Purple card for police personnel
4. **Enter Credentials:** Input provided username and temporary password
5. **Password Change:** System will prompt for new secure password
6. **Profile Setup:** Complete profile information (phone, location, etc.)

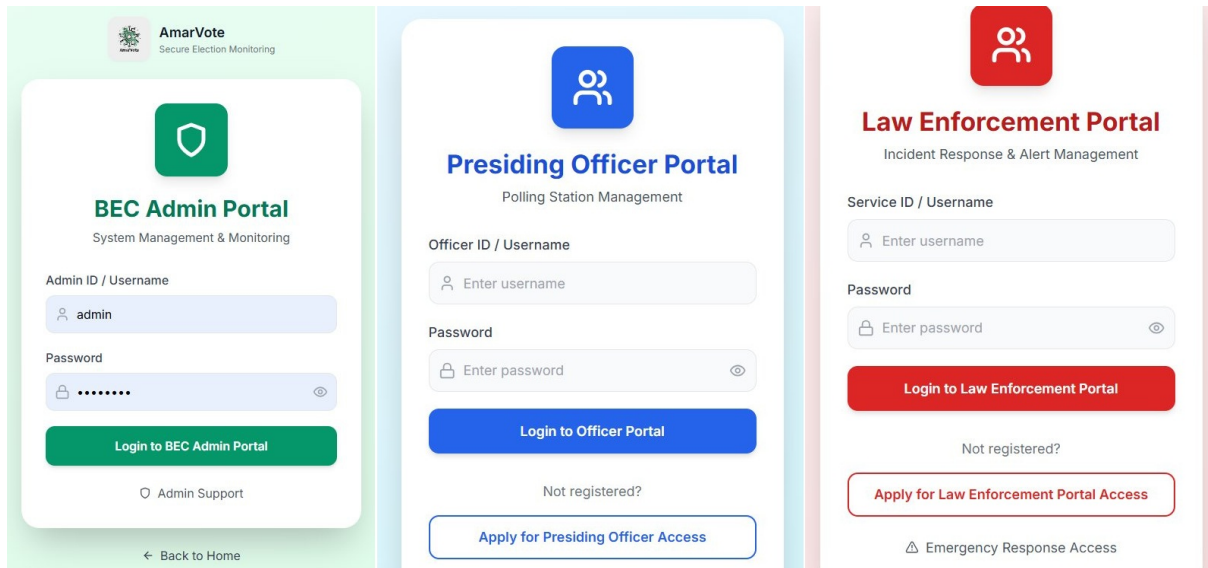


Figure 2: Role-Based Login Portals – BEC Admin, Presiding Officer, Law Enforcement

Regular Login Process:

1. Navigate to AmarVote URL
2. Select your role from the homepage
3. Enter username and password

3.2 Logout Process

Standard Logout:

1. Click user profile icon (top-right corner)
2. Select “Logout” from dropdown menu
3. Confirm logout in dialog box
4. System redirects to homepage
5. Session data cleared from browser

Automatic Logout Scenarios:

- Browser/tab closure (optional setting)
- Multiple concurrent logins detected
- Administrative session termination

4 User Roles

4.1 Role Hierarchy and Permissions Matrix

4.1.1 BEC Admin (Administrator)

Primary Responsibilities: System oversight, user management, and comprehensive monitoring.

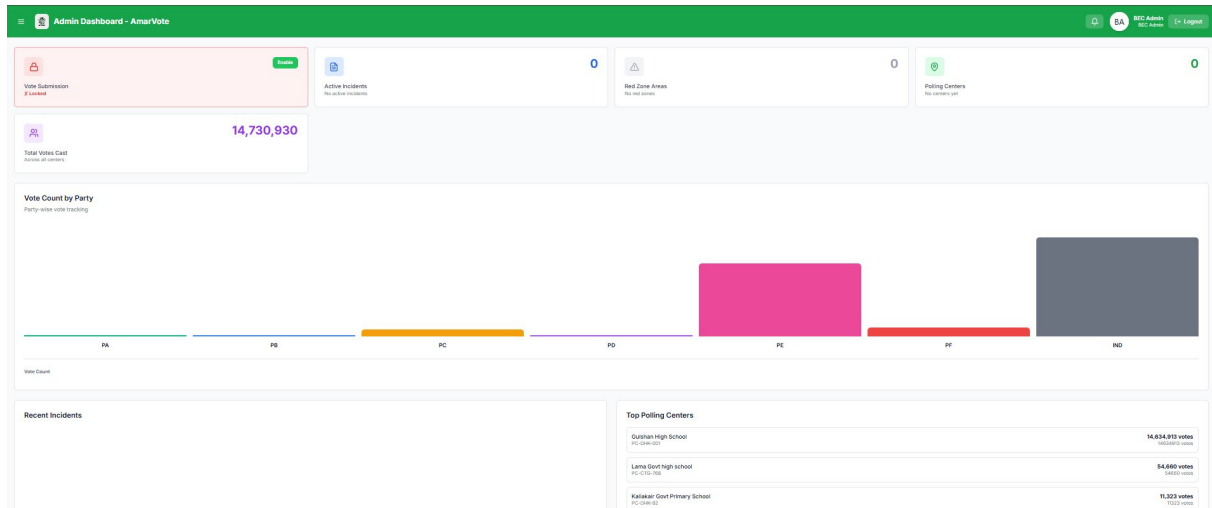


Figure 3: BEC Admin Dashboard

Permissions (Can Do):

- View all system data across all locations
- Manage user accounts (create, modify, deactivate)
- Configure system settings and policies
- Approve vote correction requests
- Generate comprehensive reports and analytics
- Access complete audit logs
- Manage polling center assignments
- Configure automated alerts and notifications
- Export data in multiple formats (PDF, Excel, CSV)
- Monitor system performance and security

Restrictions (Cannot Do):

- Submit votes directly (must be done by officers)
- Modify submitted votes without proper workflow
- Access the system without proper authentication
- Delegate admin privileges without approval

4.1.2 Presiding Officer

Primary Responsibilities: Vote submission, incident reporting from assigned polling centers.

Permissions (Can Do):

- Submit official vote counts for assigned polling center

- Request vote corrections with justification
- Report incidents with photo and GPS evidence
- Update personal profile and contact information
- View incident history for their location
- Access voting submission history
- Upload supporting documents and photos

Restrictions (Cannot Do):

- Access data from other polling centers
- Manage other user accounts
- Modify system configuration
- Access comprehensive system reports
- View audit logs beyond their activities
- Submit votes multiple times without approval

4.1.3 Law Enforcement (Police)

Primary Responsibilities: Incident monitoring, response coordination, and security oversight.

Permissions (Can Do):

- View incidents in assigned jurisdiction
- Update incident status and investigation notes
- Create new incident reports
- Access interactive map with incident locations
- Upload response photos and documentation
- Coordinate with other law enforcement units
- Generate incident response reports

Restrictions (Cannot Do):

- Access voting data or vote counts
- Manage user accounts or system settings
- View incidents outside their jurisdiction
- Modify vote-related information
- Access admin-level system functions

5 Features and Functionality**5.1 BEC Admin Features****5.1.1 User Management**

Purpose: Manage all system users and their permissions.

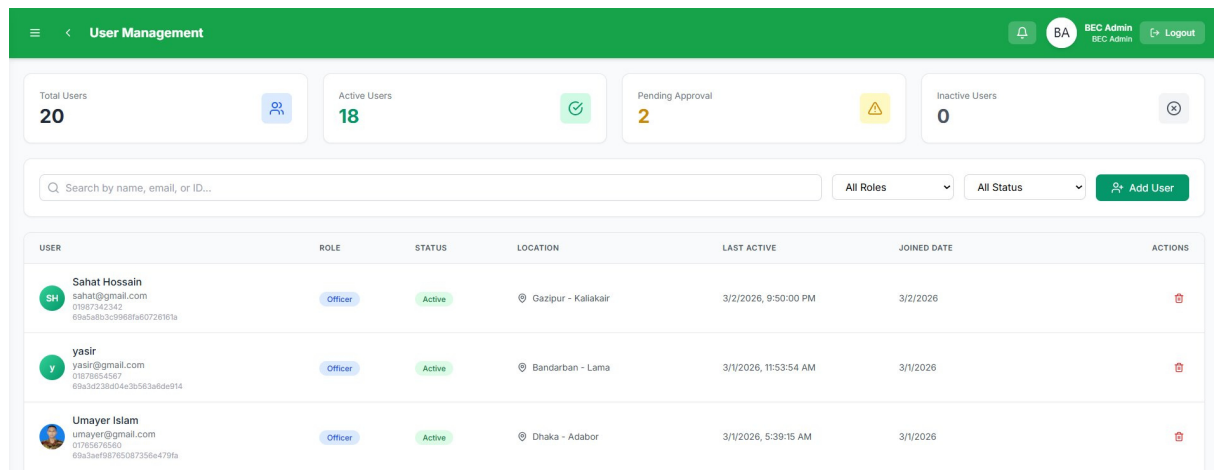


Figure 4: User Management Interface

How to Use:

1. Navigate to User Management from sidebar
2. View Users: See complete list with filters
3. Add User: Click “Add User” → Fill form → Submit
4. Edit User: Click user name → Modify details → Save
5. Manage Status: Toggle Active/Inactive status

Input and Output:

- **Input:** New or updated user details (name, role, polling center, contact, status).
- **Output:** User records created or updated, with confirmation message and updated user list.

5.1.2 Vote Analytics

Purpose: Monitor real-time vote submissions and trends.

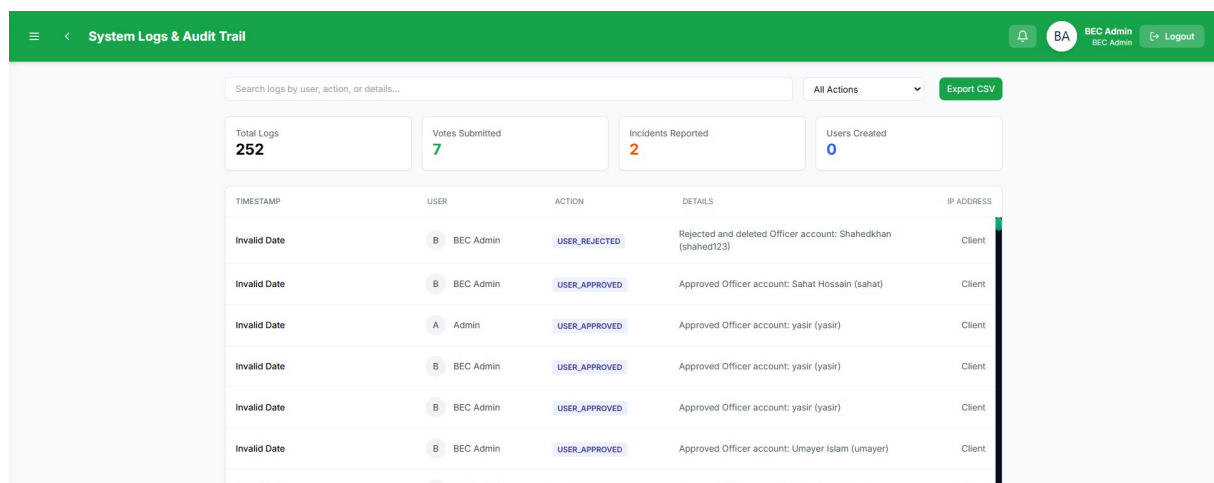


Figure 5: System Logs & Voting Trends and Analytics

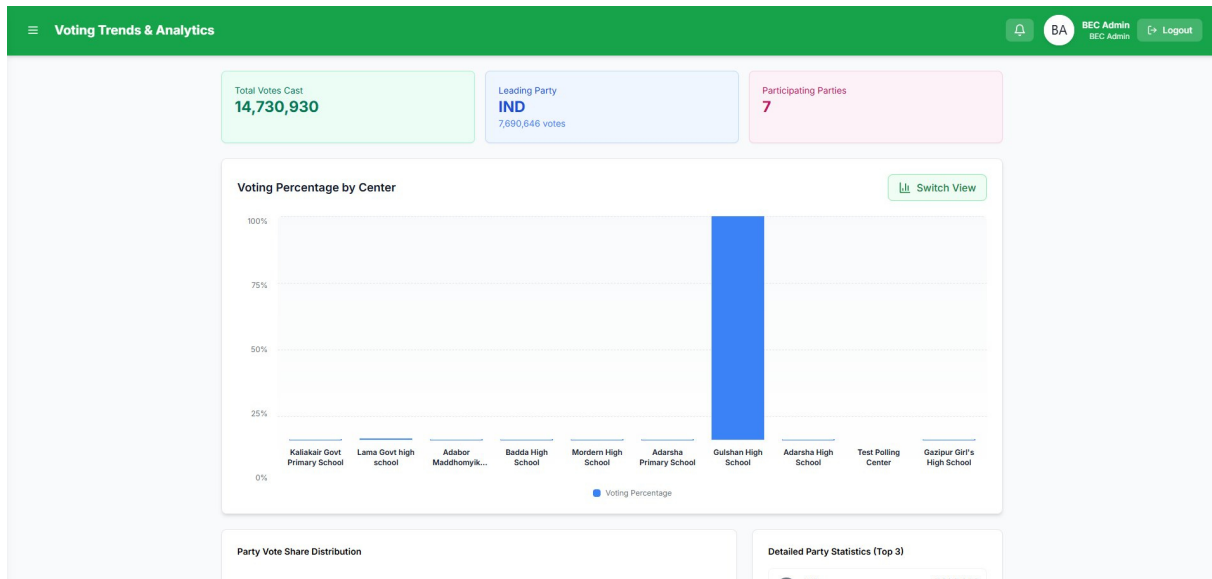


Figure 6: Voting Trends and Analytics Dashboard

How to Use:

1. Go to Voting Trends section
2. View live charts showing:
 - Total votes by polling center
 - Party-wise vote breakdown
 - Submission timeline
3. Use filters for specific locations or time periods
4. Export data as PDF or Excel reports

Input and Output:

- **Input:** Selected filters such as polling center, district, political party, and time range.
- **Output:** Updated charts and tables reflecting filtered vote data, plus downloadable report files.

5.1.3 Incident Oversight

Purpose: Monitor all incidents across all locations.

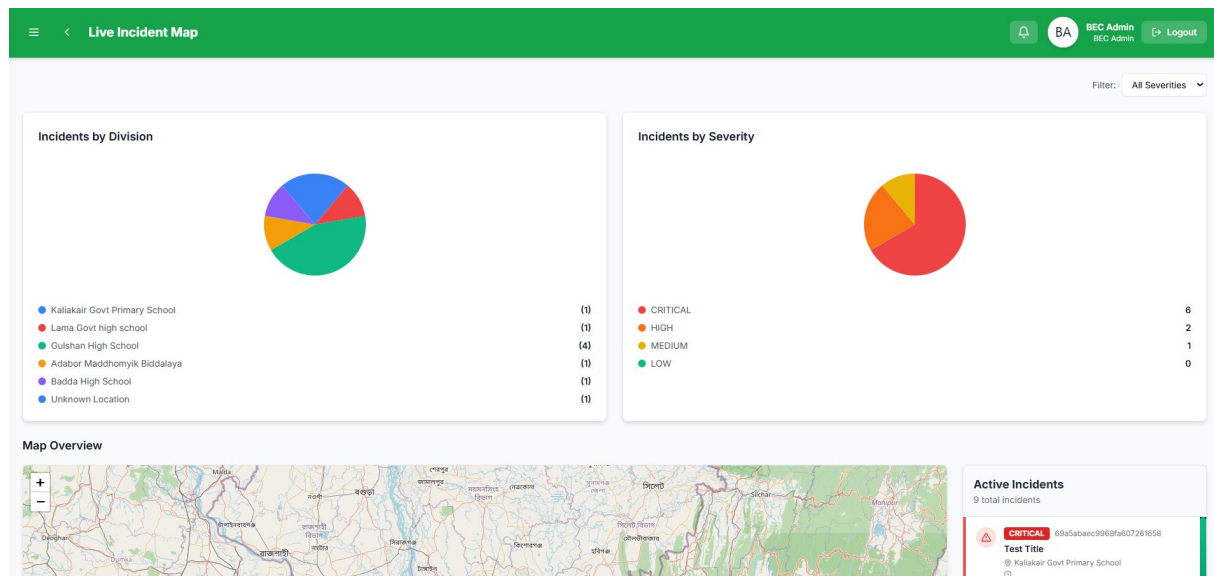


Figure 7: Live Incident Map – Incident Oversight Dashboard

How to Use:

1. Navigate to Incidents section
2. Filter by severity, status, or location
3. Click incident title to view full details
4. Review photos, GPS coordinates, and responses
5. Track resolution progress

Input and Output:

- **Input:** Selected filters (severity, status, location) and any admin comments.
- **Output:** Filtered incident list, detailed incident view, and updated status history.

5.2 Presiding Officer Features

5.2.1 Vote Submission

Purpose: Submit official vote counts from your polling center.

How to Use:

1. Go to Vote Submission section
2. Enter vote counts for each political party:
 - Awami League
 - Bangladesh Nationalist Party (BNP)
 - Jatiya Party
 - Other registered parties
3. Review all entries carefully
4. Click Submit Votes
5. Save confirmation receipt

Presiding Officer Dashboard - AmarVote

Voting In Progress
Vote entry will be unlocked when voting period ends

Party-wise Vote Entry

Vote entry is locked until voting period ends
Please wait for official voting period to end

Polling Center ID
PC-DHK-001

Total Voters in This Center
0

Enter Vote Counts by Party

PA Party A	0
PB Party B	0
PC Party C	0
PD Party D	0

Figure 8: Presiding Officer – Vote Submission Page

Input and Output:

- **Input:** Vote counts for each political party and, if required, total registered and total votes cast.
- **Output:** Stored official vote record for the polling center and a submission ID/confirmation message.

Note: Vote submission is one-time only. Contact admin for corrections.

5.2.2 Vote Correction Request

Purpose: Request permission to correct submitted votes.

How to Use:

1. Navigate to Vote Correction section
2. Click Request Correction
3. Provide detailed justification
4. Submit request and wait for admin approval
5. Once approved, resubmit with correct numbers

Input and Output:

- **Input:** Polling center selection (if applicable), correction request text, and corrected vote counts.
- **Output:** Logged correction request with status (Pending/Approved/Rejected) and audit trail.

5.2.3 Incident Reporting

Purpose: Report any irregularities or issues at your polling center.

How to Use:

1. Click Report Incident button
2. Fill incident form:
 - **Title:** Brief description
 - **Description:** Detailed explanation
 - **Severity:** Critical / High / Medium / Low
 - **Photos:** Upload evidence (max 5MB each)
3. Location is auto-detected via GPS
4. Submit and receive tracking number

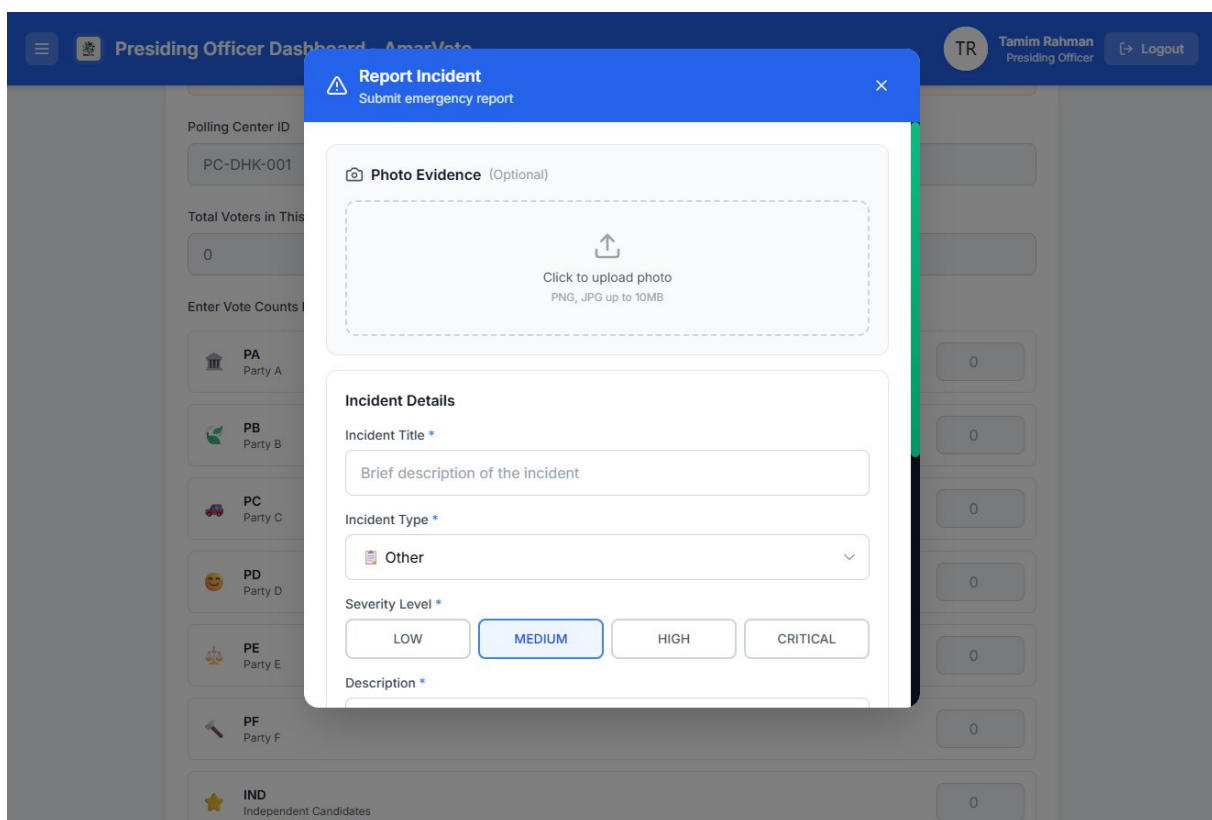
The image shows a screenshot of the AmarVote Presiding Officer Dashboard. A modal window titled 'Report Incident' is open, allowing a user to submit an emergency report. The dashboard background shows a sidebar with various polling center options (PA Party A, PB Party B, PC Party C, PD Party D, PE Party E, PF Party F, IND Independent Candidates) and a main area with input fields for 'Polling Center ID', 'Total Voters in This', and 'Enter Vote Counts'. The 'Report Incident' modal includes a 'Photo Evidence' section with an upload button, an 'Incident Details' section with fields for 'Incident Title', 'Incident Type', 'Severity Level', and 'Description', and a 'Submit' button. The user's name 'Tamim Rahman' and a 'Logout' button are visible in the top right corner of the dashboard.

Figure 9: Incident Reporting Form

Input and Output:

- **Input:** Incident title, description, severity level, photos, and automatically captured GPS location.
- **Output:** New incident record with unique tracking ID and notifications to relevant stakeholders.

5.3 Law Enforcement Features

5.3.1 Incident Monitoring

Purpose: Track and respond to incidents in your jurisdiction.

How to Use:

1. View Incidents dashboard
2. Filter by jurisdiction and severity
3. Prioritize Critical and High severity incidents
4. Click incident to view details and evidence
5. Update status as investigation progresses

Input and Output:

- **Input:** Selected jurisdiction, severity filters, and any updated status or notes.
- **Output:** Updated incident list and detailed views reflecting police actions and current status.

5.3.2 Incident Response

Purpose: Update incident status and document response actions.

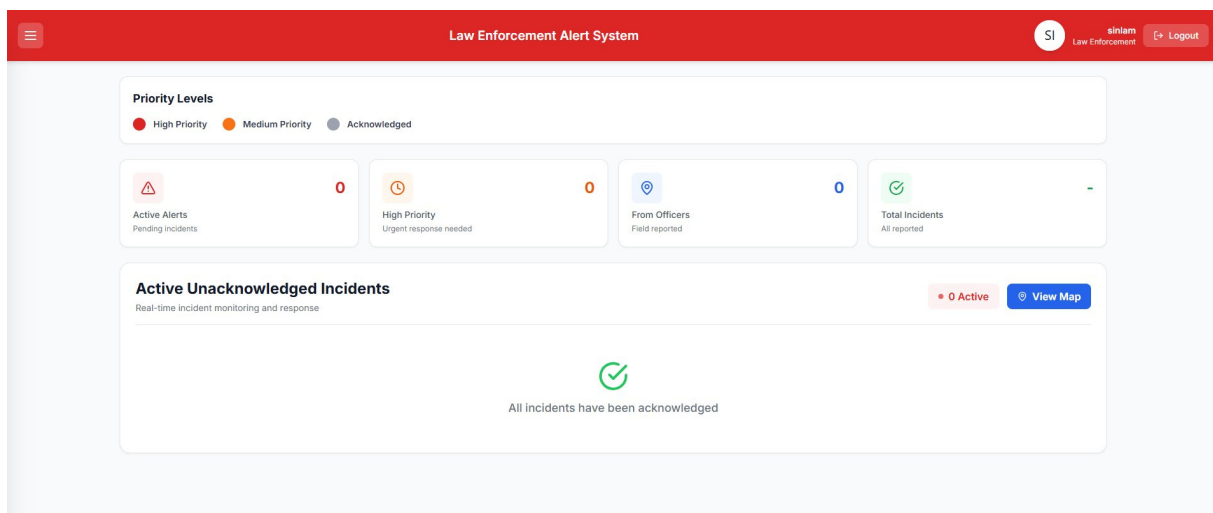


Figure 10: Law Enforcement Alert and Monitoring System

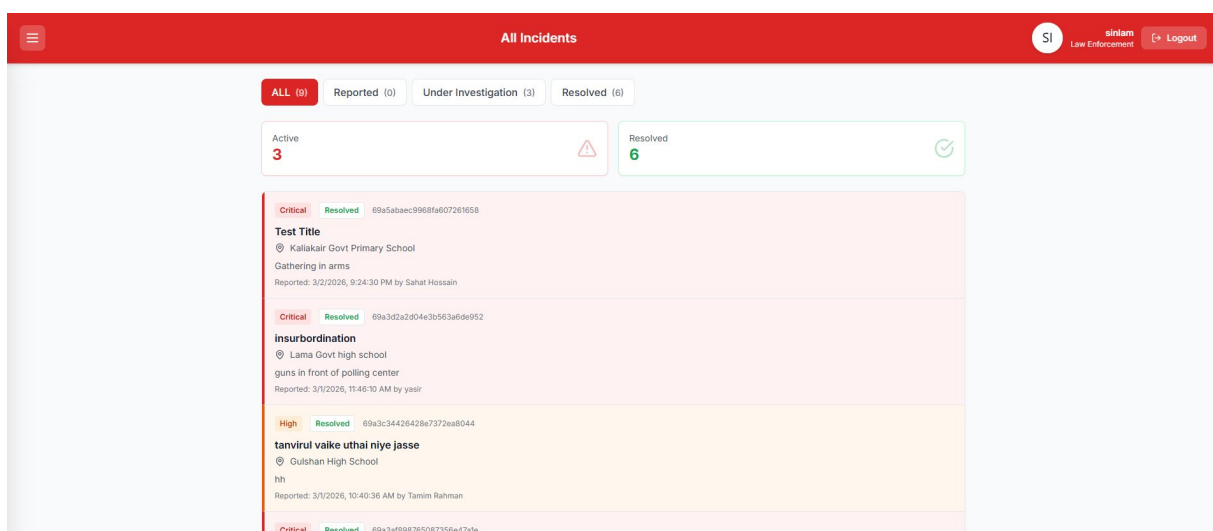


Figure 11: All Incidents – Law Enforcement View

How to Use:

1. Open incident details page
2. Update Status:
 - Under Investigation
 - Resolved
 - Dismissed
3. Add Response Notes describing actions taken
4. Upload photos of response activities
5. Submit updates for record keeping

Input and Output:

- **Input:** Selected incident, new status, response notes, and any response photos.
- **Output:** Updated incident record with new status, attached evidence, and full response history.

5.3.3 Interactive Map

Purpose: Geographic view of incidents and polling centers.

How to Use:

1. Navigate to Map section
2. View all incidents and polling centers on map
3. Click markers for detailed information
4. Use filters to show specific types
5. Plan efficient response routes

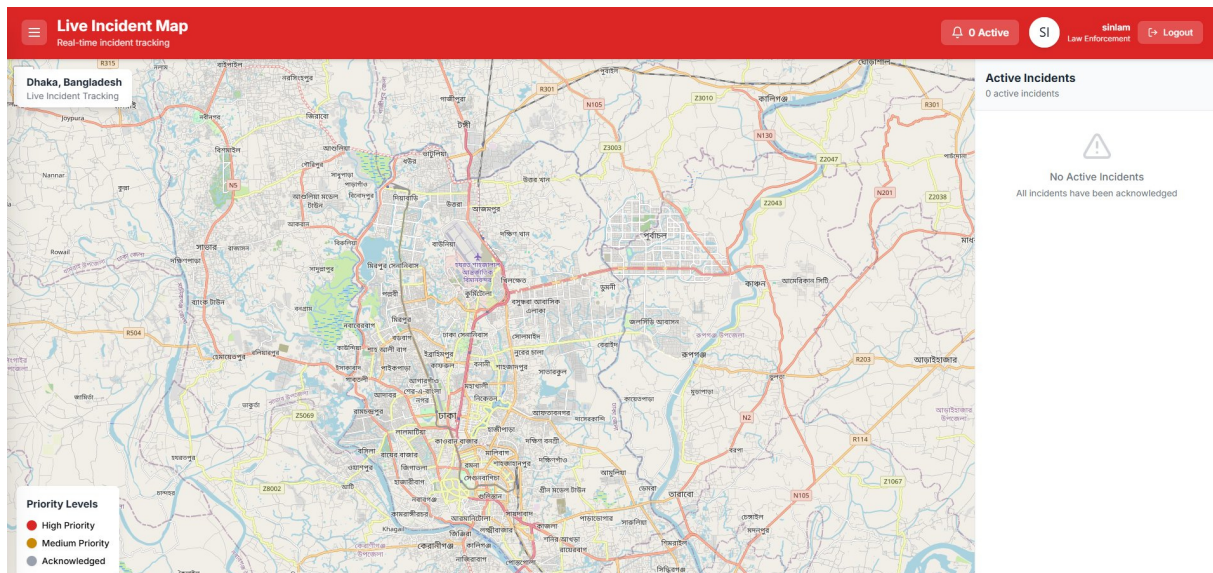


Figure 12: Law Enforcement Interactive Map – Live Incident Tracking

Input and Output:

- **Input:** Selected filters (incident type, severity, status) and map navigation actions.

- **Output:** Visual map view of incidents and polling centers with contextual details per marker.

6 Navigation Guide

6.1 Menu Structure and Dashboards

Role-Based Menus:

- **BEC Admin Dashboard:** Sidebar with links to User Management, Polling Centers, Voting Trends, Incidents, Audit Logs, and System Settings.
- **Presiding Officer Dashboard:** Sidebar with Vote Submission, Vote Correction, Incidents, and Profile.
- **Law Enforcement Dashboard:** Sidebar with Incidents, Map, Reports, and Profile.

Each dashboard has a top bar showing the logged-in user, role, and quick actions such as notifications and logout.

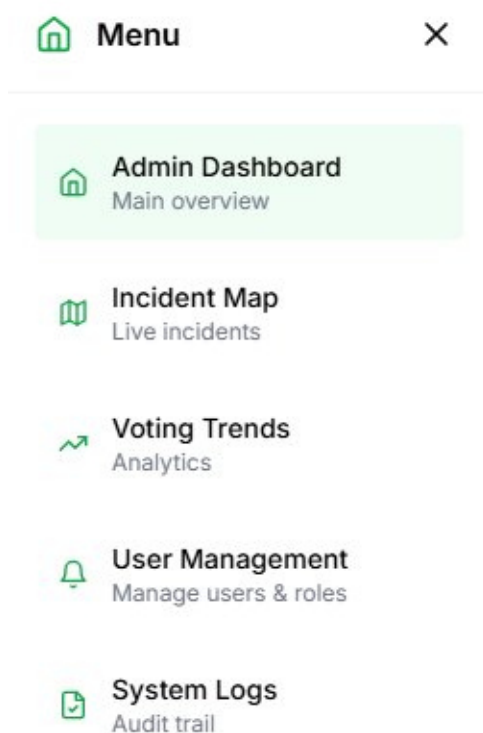


Figure 13: Navigation Menu Structure

6.2 Page Descriptions

- **Home / Dashboard:** Summary cards for total incidents, vote submissions, and key alerts.
- **User Management (Admin):** Tabular list of users with search, filters, and action buttons (edit, deactivate).
- **Voting Trends (Admin):** Charts and graphs displaying vote counts and party-wise breakdowns.

- **Vote Submission (Officer):** Form-based page for entering vote counts and submitting results.
- **Incidents (Officer/Police):** List of incidents with status, severity, and quick links to details.
- **Map (Police):** Full-screen map view showing incident and polling center markers.

6.3 Buttons, Icons, and Special Elements

- **Primary Buttons:** Solid colored buttons (e.g., “Submit”, “Save”) used for main actions.
- **Secondary Buttons:** Outlined buttons for less critical actions (e.g., “Cancel”, “Back”).
- **Status Badges:** Colored labels indicating incident status (e.g., Critical, High, Medium, Low, Resolved).
- **Notification Bell:** Icon in the header showing new alerts when highlighted.
- **Profile Avatar:** User icon in the header; clicking opens profile and logout options.

7 Error Handling and Messages

7.1 Common Errors and Causes

Frequently Encountered Issues:

- **Login failed:** Incorrect username or password, locked account, or expired session.
- **Vote submission blocked:** Votes already submitted for the polling center or missing required fields.
- **Photo upload failed:** File size too large, unsupported format, or unstable internet connection.
- **GPS not available:** Browser location permissions denied or device GPS turned off.
- **Slow system response:** Poor network connectivity or heavy load during peak hours.

7.2 Error Messages

Error	Meaning and Solution
“Invalid credentials”	Username or password is incorrect. Re-enter carefully or contact admin if you forgot them.
“Invalid password”	The password does not meet the required criteria (length, complexity, or character types). Choose a stronger password.
“Rate limit exceeded”	Too many login attempts. Wait 15 minutes before trying again.
“Votes already submitted”	Votes for this polling center are already recorded. Submit a vote correction request if needed.
“Photo upload failed”	Image too large or unsupported format. Use JPG/PNG under 5MB and retry.
“Token expired”	Session has expired. Log out (if possible) and log in again.

Table 1: Common System Error Messages

7.3 How to Fix Common Issues

7.3.1 Login Issues

Problem: Cannot access the system

1. Verify your internet connection is stable.
2. Confirm you are using a supported browser and latest version.
3. Check that the URL is correct and from an official source.
4. Re-enter your username and password carefully.
5. If the account may be locked or suspended, contact your administrator.

7.3.2 Performance Issues

Problem: System running slowly

1. Test your internet speed (minimum 1 Mbps, recommended 5 Mbps).
2. Close unnecessary browser tabs and applications.
3. Clear browser cache and try again.
4. If possible, switch to another supported browser.
5. Report to technical support if the issue persists during normal network conditions.

7.3.3 Data Sync Issues

Problem: Information not updating

1. Refresh the page (F5 or Ctrl+R).
2. Confirm internet connection stability.
3. Log out and log in again to refresh your session.
4. Contact support if critical data (votes or incidents) appears incorrect.

7.3.4 Mobile Device Issues

Problem: System not working properly on mobile

1. Use the latest version of Chrome (Android) or Safari (iOS).
2. Ensure JavaScript is enabled in browser settings.
3. Allow location access for incident reporting when prompted.